

Bedaya MVP2: Project Documentation

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Overview

Bedaya (بداية) is one bilingual website where a person in Jordan signs in with SANAD and gets everything needed to open a business, guided by Saad (سعد), an AI copilot that prepares the work while the owner approves each step.

The problem. Opening even a small business in Jordan means visiting several offices (the Ministry of Industry, Trade and Supply, the chambers, the municipality, JFDA, the tax department, Social Security). Fees, papers and the order of steps are scattered across sources, and first-time founders lose weeks to repeat visits and missing documents.

What Bedaya does.

- Builds a personal roadmap with every step, office, paper, fee and timing, taken from official Jordanian sources.
- Fills the government forms from the owner's verified SANAD identity and lets them sign with SANAD in one place.
- Matches the business with real incubators, grants and loans, and prepares a bank file with only the documents the owner selects.
- Gives the owner Saad: an AI copilot that writes the brief, the plan and the paperwork, recommends the right legal structure, and answers questions in Arabic or English.
- Helps the business look real from day one: brand kit, designs and a published website (Launch Studio), plus HR, payroll, domain, accounting and hiring tools.

MVP2 is the release published on GitHub as the `main` and `MVP2` branches of [Beluga_HTU](#). It runs fully on a laptop: the AI uses a free open-source model locally, with no API key and no data leaving the machine.

Who uses it

Five kinds of account each see only their own pages; a bank, incubator, expert or admin opening an owner page is sent back to their own home.

Party	Lands on	What they can do
Business owner	Saad (the AI copilot)	Choose a legal structure, approve the brief, plan and paperwork, follow the roadmap, upload and check documents, sign forms with SANAD, apply for funding, send a bank file, book experts, build a brand and website, run HR, payroll and accounting
Bank	Bank inbox	Review the bank files sent to it (with only the documents the owner selected) and record a decision the owner sees
Incubator	Incubator inbox	Review applications to its programme and approve, reject or request documents
Expert	Their availability	Publish free time slots, see bookings, cancel; a slot can never be double-booked
Admin	Management	Edit the rules catalogue and roles, see analytics, and act as each government office on the staff dashboards to approve or return signed forms

Sign-in works two ways: **Login with SANAD** (national ID and SANAD password, then consent) opens the account linked to that national ID; **email and password** works for every party. A SANAD identity with no account can create one in a single step: SANAD fills the name, national ID, birth date, phone and address, marked as verified, and the person adds only an email and password.

The owner journey and features

An owner signs in, answers nine questions once, and from then on works through eight menu sections; every page has a breadcrumb and a "Back to main page" button, and the whole site works in Arabic (right to left) and English, on phone and desktop.

1. **Sign in with SANAD.** A three-step screen: national ID and SANAD password, consent listing exactly what Bedaya receives (identity, contact, address), then back to Bedaya.
2. **Onboarding.** Nine questions (activity, city, premises, name, partners, capital, staff, support needed, details). Answers prefill from SANAD and build the profile and the roadmap.

3. Saad, the AI copilot. The home page after sign-in: legal structure, business brief, launch plan, documents, then signing, each approved by the owner (next section).

Menu section	Pages inside (tabs)	What it does
Saad	One page, four stages	AI copilot with human approval at every step
Your roadmap	Steps, Business plan, Location, Compliance	Ordered steps with office, papers, fee and time; a business plan PDF in Arabic or English; a map to place the premises and check them against zoning areas; post-launch obligations
Documents & signing	Documents, Sign forms	Upload papers with automatic quality and expiry checks; government forms filled from the profile and signed with SANAD in one go, then sent to each office
Funding & support	Incubators, Funding, Bank file, My applications, Experts, Appointments	Real programmes matched with reasons; a bank file that sends only the documents the owner ticks; expert booking from each expert's own availability
Launch Studio	Overview, Brand kit, Designs, Website	Names, palette with contrast check, fonts, four logos and a brand guide PDF; social posts, business card, flyer, menu and shop sign; a bilingual one-page website published to a subdomain
Business tools	HR & payroll, Domain & email, Accounting, E-invoicing, Online presence, Hiring	Contracts and leave; payroll with Social Security and income tax; domain and email DNS records; branded invoices; Google Maps and social texts; job ads and CV ranking

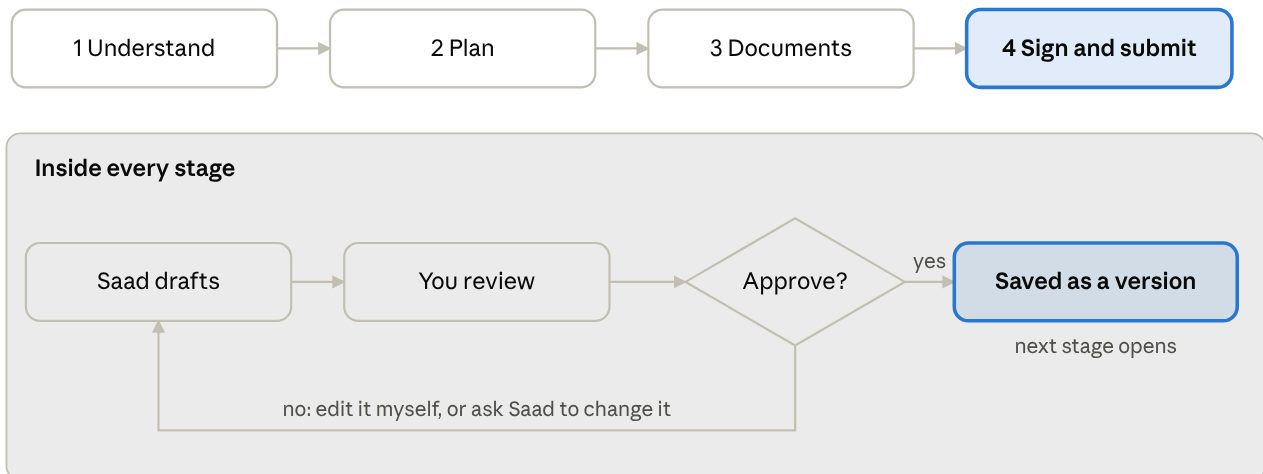
Menu section	Pages inside (tabs)	What it does
Assistant	Chat with saved past conversations	Ask Saad anything, in Arabic or English
Notifications	One page	Updates from the roadmap, bookings and government offices

The Saad character also floats in the corner of every page ("Ask Saad"), opening the same chat without leaving the page.

Saad, the AI

Saad drafts and the owner decides: nothing the AI writes counts until the owner approves it, and every draft, edit and approval is saved as a version and listed in an activity log marked "AI" or "You".

Saad drafts, you decide: each stage opens after you approve



human in the loop · 4 stages, one approval loop inside each

A draft only moves on when the owner approves it; editing or asking Saad for a change sends it round the loop again.

The copilot's four stages. Each stage opens only after the one before it is approved.

- 1. Understand.** First the legal structure: Saad compares home business, sole proprietorship (مؤسسة فردية) and LLC (ذات مسؤولية محدودة), with owners, liability, official fees, steps and time computed from the official rules for this owner, when to choose each, and what to watch out for. He recommends one with reasons drawn from

the business (working from home, partners, capital, hiring, sector). General partnership, limited partnership, private shareholding and non-profit companies are explained too, with "Ask an expert". Switching rebuilds the roadmap, and is refused once a step is done. Then Saad writes a one-page business brief.

2. **Plan.** Every step says where to go, how (sign with SANAD in Bedaya, in person, or at a shop), what to sign, the papers, the fee, the time and a tip, plus Saad's recommendations for this kind of business. The owner approves or skips each step.
3. **Documents.** Saad writes the paperwork an AI can draft (activity description, trade-name options, the landlord's approval letter, home-business commitments, a food-authority product sheet, a funding letter); Bedaya fills the official forms. The owner reviews, edits or asks for changes, approves each one, and downloads branded PDFs.
4. **Sign and submit.** Only approved forms go on to SANAD signing.

On every draft the owner has three choices: **Approve**, **Edit myself**, or **Ask AI to change** (for example "make it shorter").

The chatbot. The same Saad answers questions on the Assistant page and from the floating character. He greets the owner by name with their progress and suggests questions from their next step. Answers stream word by word, include buttons to the right Bedaya page and the official sources, and past conversations are saved to the account.

How the answers stay right.

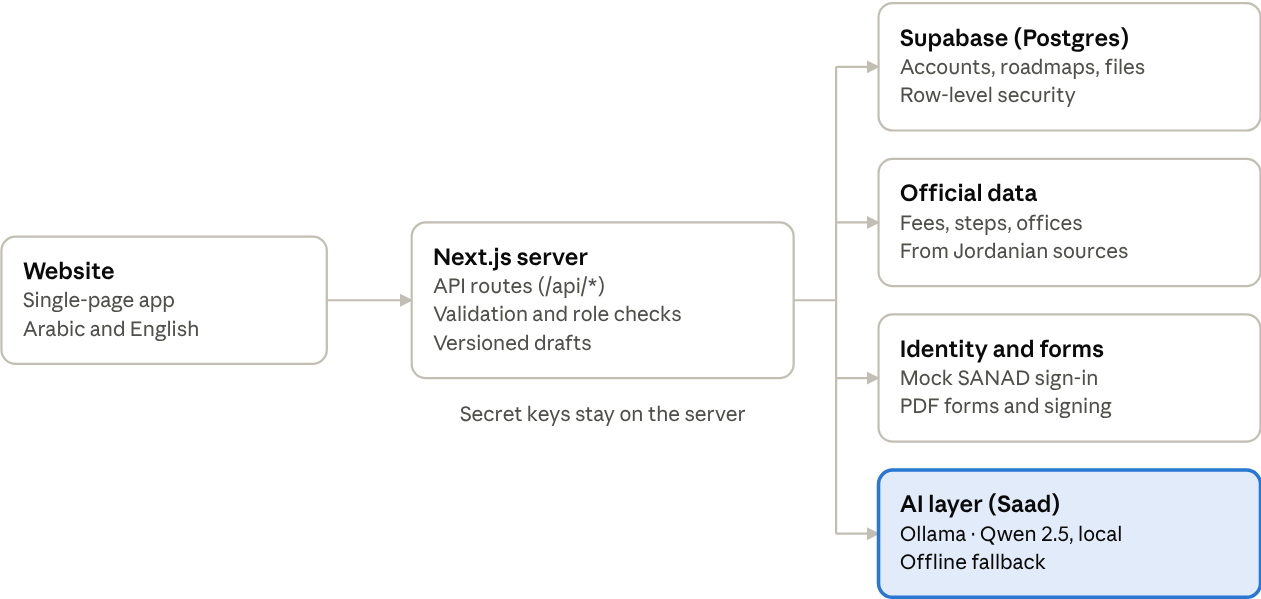
- **Grounded, not trained.** For each question, Bedaya retrieves the matching official facts (fees, papers, offices, timings) and reads the owner's own situation on the server (business, roadmap progress, next step, missing papers, total fees, funding matches). The model writes the answer from that context only; nothing is fine-tuned, so updating the data updates Saad.
- **Exact numbers.** When Bedaya can compute the answer (total fees, next step, missing papers, funding), the model only rephrases that verified answer and keeps every number.
- **On topic.** Requests that are not about a business in Jordan (a poem, sport) get a polite refusal without calling the model.
- **Safe output.** Model text is escaped before display; only bold, lists and links to known Bedaya pages are rendered. Websites are JSON that fills Bedaya's own template, and AI-made logos pass an allow-list SVG sanitiser.
- **Always available.** If the model is off or slow, offline answers built from the same data take over.

The model. Saad runs Qwen 2.5 7B, an open-source model, through Ollama on the laptop's GPU: free, no API key, no internet, and no data leaves the machine. The first words appear in about 1.5 seconds; a full chat answer takes 5 to 9 seconds and a copilot draft 15 to 30 seconds. The same layer can switch to Claude through the Anthropic API with one setting.

Architecture and technology

Bedaya is one Next.js 16 application: a single-page website in the browser, API routes on the server, Supabase for accounts and data, and an AI layer that talks to a local model.

One Next.js app: the browser talks only to its own server



architecture · one app, four back-end services

The browser only ever calls Bedaya's own API. The server holds the keys and talks to the database, the official data, identity and forms, and the local AI model.

Layer	Technology	Where in the code
Website	Plain ES-module JavaScript single-page app with hash routes, Lucide icons, Leaflet maps, pdf-lib for PDFs	<code>public/bedaya/js/ (main , core , onboarding , owner , partners , tools , staff , studio , services , chat , copilot) , app/site.css</code>

Layer	Technology	Where in the code
Server	Next.js 16 App Router, React 19, TypeScript 5.9, Zod validation	<code>app/api/*</code> , <code>lib/*</code>
Accounts and data	Supabase (Postgres with row-level security, security-definer functions, private file storage)	<code>supabase/migrations/</code> , <code>lib/supabase/</code> , <code>lib/account.ts</code>
Identity, forms and signing	Mock SANAD behind an <code>IdentityProvider</code> interface; PDF forms filled from verified data; office review	<code>lib/integrations/{sanad,ocr,pdf,documents}</code> , <code>app/mock-sanad</code> , <code>app/dashboard</code>
AI	One provider layer: Ollama (local), Claude, or offline answers	<code>lib/integrations/llm.ts</code> , <code>lib/integrations/assistant/</code> , <code>lib/copilot.ts</code> , <code>lib/legal-forms.ts</code>
Published sites	Owner websites served at <code>/s/<name></code> and <code><name>.<domain></code>	<code>app/s/[slug]</code> , <code>proxy.ts</code> , <code>lib/studio/</code>
Official data	Rules, offices, documents, programmes and their sources	<code>data/m1/</code> , <code>lib/integrations/knowledge.json</code>

Main API groups. `/api/account` (who is signed in, SANAD login and sign-up), `/api/platform` (business, onboarding, roadmap, documents, applications, bookings, partner and admin), `/api/copilot` (context, brief, plan, documents, legal structures), `/api/ai/chat` (streaming chat, welcome, saved conversations), `/api/experts` , `/api/workspace` (versioned drafts and approvals), `/api/studio` , `/api/services` , `/api/location` .

Versioned drafts. Everything the owner approves (Saad's brief, plan and documents, the brand kit, designs, the website, HR records) is stored as numbered versions with an approved flag, so any earlier version can be reopened.

Team. Member 1 prepared the official data and rules, Member 2 the interface design, Member 3 the frontend behaviour, Member 4 the backend and Supabase, and Member 5 (Ameen and Abdullah) the AI, identity, documents and signing; the integration joined them into this one app.

Data, security and privacy

Every fee, step, office and required paper comes from official Jordanian sources and keeps its source link; the parts that depend on government systems are simulated and labelled.

Real	Simulated for the MVP
Fees, steps, times, offices and papers from MIT eRegulations, Greater Amman Municipality, JFDA, the chambers, the Income and Sales Tax Department and Social Security (<code>data/m1/sources.json</code>)	SANAD sign-in and e-signatures (the SANAD screens say "demo")
Support programmes: JEDCO, the Development and Employment Fund, Orange Corners, Oasis500, QRCE, iPARK, Luminus	Document reading (OCR)
Social Security rates (7.5% employee, 14.25% employer) and income-tax bands with the 9,000 JOD exemption	Government form templates, stamped "DEMO FORM - NOT OFFICIAL"
The open-source AI model	The partner bank, appointment slots, zoning areas, domain purchase, site hosting, and email or WhatsApp sending (logged only)

Security and privacy.

- **Each account sees only its own data.** Supabase row-level security on every table; owner pages are closed to other parties; staff dashboards open only for the admin account, and the staff role is re-checked against the account on every request.
- **One identity per account.** A SANAD national ID is linked to exactly one account; signing in always binds the session to that account's own identity, and signing out clears every session and the browser's stored data.

- **Keys stay on the server.** All keys are environment variables, never in code; the Supabase secret key is server-only and used only to open the account linked to a verified SANAD login.
- **Consent is explicit.** SANAD shows exactly what Bedaya receives; sharing a file with a programme or bank needs the owner's confirmation, and the bank receives only the documents the owner ticks.
- **AI is private by default.** The model runs locally; sending data to an external AI requires an explicit switch (`BEDAYA_ALLOW_EXTERNAL_AI=true`).
- **Safe content.** Everything shown is escaped; uploads are checked by type, size and file signature; AI output never becomes HTML.

Setup and running

The app needs Node 20 or later, a Supabase project (or a local one in Docker) and, optionally, Ollama for the AI; without Ollama every AI feature falls back to offline answers.

1. **Get the code:** `git clone https://github.com/Amin-Shahin-00/Beluga_HTU.git` , then `npm install` .
2. **Database:** in the Supabase SQL Editor, paste and run `supabase/setup-all.sql` . It creates every table and function and adds the demo accounts, and is safe to run again (`npm run db:bundle` rebuilds it after a migration changes).
3. **Settings:** copy `.env.example` to `.env.local` and fill in the values below.
4. **AI (optional):** install Ollama (`winget install Ollama.Ollama`) and run `ollama pull qwen2.5:7b` (about 4.7 GB; runs on a 6 GB GPU).
5. **Run:** `npm run build` then `npm start` (or `npm run dev` while developing), and open `http://localhost:3000`.

Setting	Value	Notes
<code>NEXT_PUBLIC_SUPABASE_URL</code>	The project URL	Safe for the browser
<code>NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY</code>	The publishable key	Safe for the browser
<code>SUPABASE_SECRET_KEY</code>	The secret key	Server only; needed for Login with SANAD
<code>LLM_PROVIDER</code>	<code>ollama</code> , <code>mock</code> or <code>anthropic</code>	<code>ollama</code> = free local model

Setting	Value	Notes
LLM_MODEL	qwen2.5:7b	Model name for the chosen provider
LLM_API_KEY , BEDAYA_ALLOW_EXTERNAL_AI	Only for anthropic	External AI is off unless set to true
BEDAYA_SITE_DOMAIN	A wildcard domain	Optional; serves published sites at <name>.<domain>

Demo accounts. The login screen never lists these.

Party	Email	Password
Business owner (Layla, home bakery in Irbid)	layla.demo@example.com	Layla2026!
Business owner (Omar)	omar.demo@example.com	Omar2026!
Bank (Partner Bank)	bank.demo@example.com	Bank2026!
Incubator (JEDCO HBB)	incubator.demo@example.com	Incubator2026!
Experts	expert.demo@example.com , expert2.demo@example.com , expert3.demo@example.com	Expert2026!
Admin	admin.demo@example.com	Admin2026!

SANAD (mock) national IDs 99900000001 (Layla), 99900000002 (Omar) and 99900000003 (Rami, no account yet, to try "Create account with SANAD") all use the SANAD password Sanad2026! .

Testing, limitations and next steps

MVP2 passes its automated checks (data references, type checking, 20 of 20 assistant questions, 10 of 10 conversation cases, the feature suite and the production build), and every flow was tested in a browser as each party, in Arabic and English, on desktop and phone.

Check	Command	Result
Official data references	<code>npm run check:data</code>	Passes
Types	<code>npm run typecheck</code>	Passes
Assistant questions	<code>npm run test:assistant</code>	20 of 20 (target 18)
Conversation follow-ups	<code>npm run test:conversation</code>	10 of 10
Checker, matching, prefill, business plan, PDFs	<code>npm run test:features</code>	Passes
Production build	<code>npm run build</code>	Passes

Known limitations.

- SANAD, OCR, e-signatures and the government form templates are simulated until the real integrations are approved.
- The step-by-step data covers home business, sole proprietorship and LLC; other structures are explained and routed to an expert.
- Municipal and chamber rules are Amman-specific; other cities show a note to confirm with the local authority.
- A 7B model on a laptop is good but not perfect: numbers come from the data, but wording can be loose, and copilot drafts take 15 to 30 seconds.